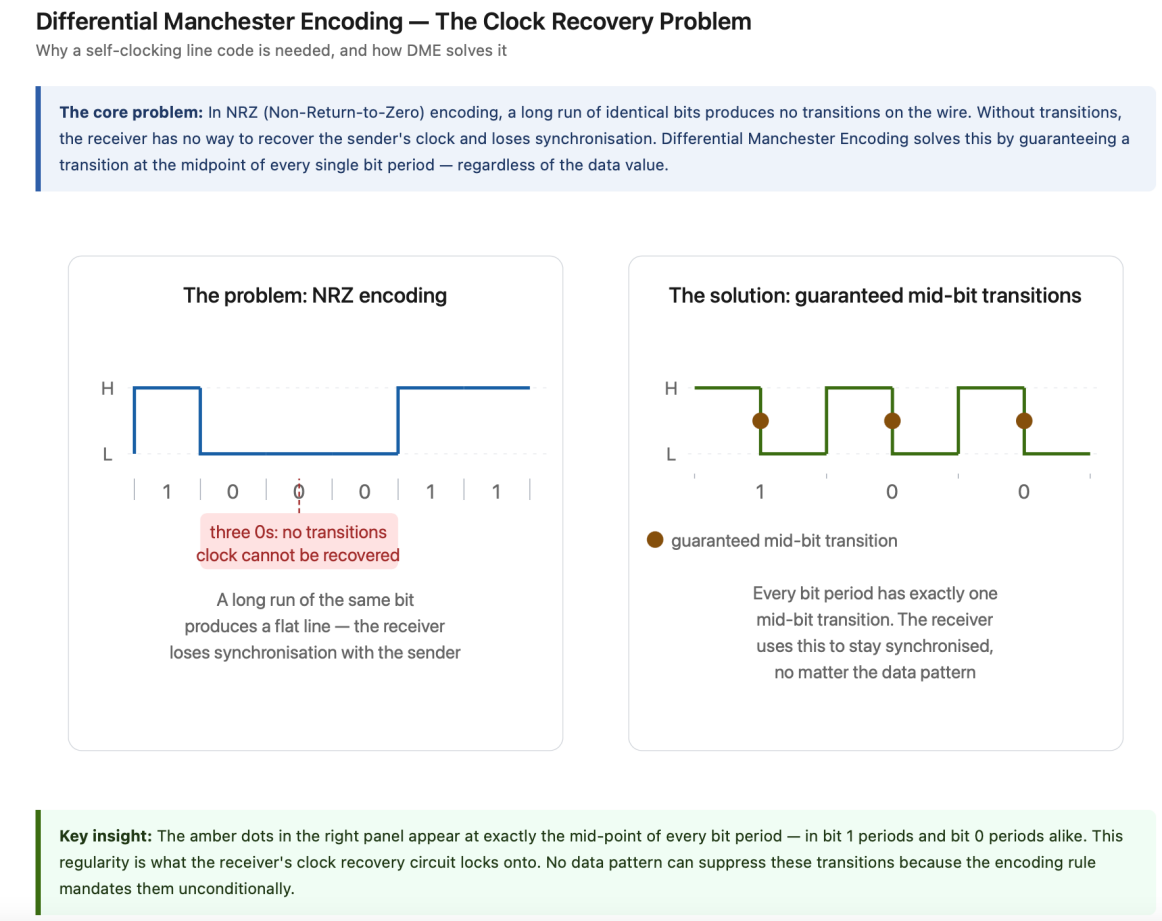
Differential Manchester Encoding — Fundamentals

The key concept: Differential Manchester is a **self-clocking, differential line code** where the clock is embedded in the signal itself and data is encoded in the **presence or absence of a mid-bit transition**, not in the signal level.

Full explanation with multiple diagrams, going from first principles through to the complete encoding.



The two rules of Differential Manchester Encoding

DME has exactly two rules, applied at every bit boundary:

Rule 1 — the mid-bit transition. Every bit period has a mandatory transition at the centre of the period. This is unconditional. It is what makes the encoding self-clocking — the receiver can always find the clock edge.

Rule 2 — the start-of-period transition encodes the bit. If the current bit is a **0**, there is an additional transition at the start of the period (at the bit boundary). If the current bit is a **1**, there is no transition at the start of the period. The signal simply continues at whatever level it was at the end of the previous bit.

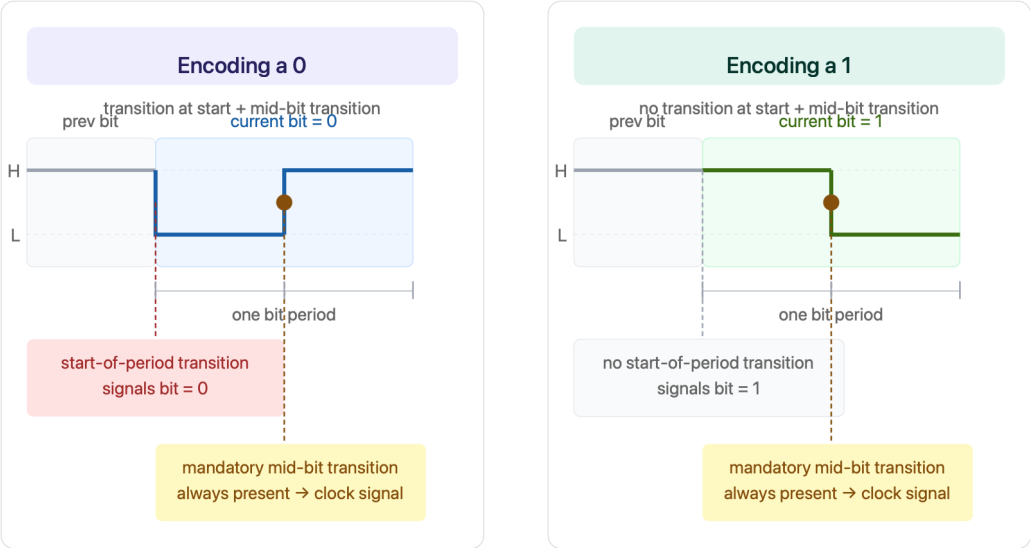
The critical insight that distinguishes *Differential* Manchester from plain Manchester is in rule 2: the bit value is encoded in the **presence or absence of a transition**, not in the

direction of the mid-bit transition. This makes it immune to polarity inversion — if the cable is wired backwards, the data is still correctly decoded.

Differential Manchester Encoding — The Two Encoding Rules

How each bit value is translated into a waveform pattern within one bit period

DME has exactly two rules applied at every bit boundary. **Rule 1** (mandatory mid-bit transition) is unconditional and provides the clock. **Rule 2** (optional start-of-period transition) carries the data: present for 0, absent for 1. Both rules together produce a waveform with 1 or 2 transitions per period.



Rule for encoding 0 — two transitions per period

At the **start** of the bit period: the signal transitions (changes level).
At the **mid-point** of the bit period: the signal transitions again (mandatory).

The receiver sees two transitions and decodes: `bit = 0`.

Rule for encoding 1 — one transition per period

At the **start** of the bit period: no transition. Signal continues unchanged.
At the **mid-point** of the bit period: the signal transitions (mandatory).

The receiver sees one transition and decodes: `bit = 1`.

Why "differential"? The bit value is encoded in the *presence or absence of a transition* at the period boundary — not in the absolute signal level (HIGH = 1, LOW = 0 style). This means if the cable is wired with reversed polarity, the transition pattern is identical and decoding is unaffected. The receiver does not need to know which level is "high" — it only needs to detect transitions.

Notice a subtle but important consequence: the **direction** of the mid-bit transition (whether it goes H→L or L→H) is not what carries the data. It simply depends on what state the signal was in. The receiver need only ask "was there a transition at the start of this period?" — yes means 0, no means 1. This is what makes the scheme differential and therefore polarity-independent.

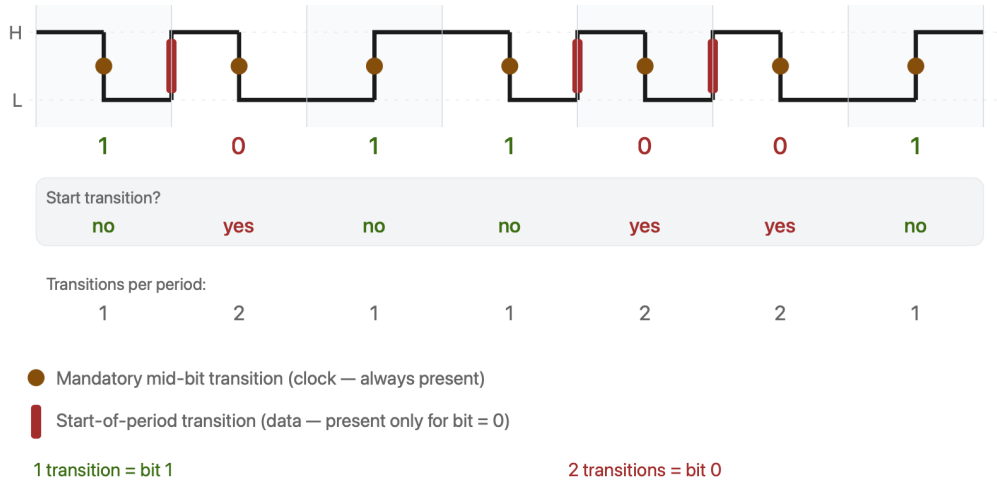
Complete worked example — encoding 1 0 1 1 0 0 1

Now watch the full sequence. Starting state: the signal is at HIGH at the beginning of bit 1.

Differential Manchester Encoding — Full Worked Example

Encoding the bit sequence **1011001** · starting state: HIGH · period width proportional to display

How to read this diagram: Amber circles mark the mandatory mid-bit transitions — present in every period, they provide the clock. Red bars mark start-of-period transitions — present only when the bit is 0, they carry the data. The decode table below shows how the receiver determines each bit by checking whether a start-of-period transition occurred.



Full DME waveform for bit sequence 1 0 1 1 0 0 1 with annotations

			Start transition?	Transitions total	Waveform description
1	1	HIGH	No — stays HIGH	1 (mid only)	Holds HIGH for half period, then HIGH→LOW at mid-point
2	0	LOW (end of period 1)	Yes — LOW→HIGH	2 (start + mid)	Jumps HIGH immediately, holds to mid-point, then HIGH→LOW
3	1	LOW (end of period 2)	No — stays LOW	1 (mid only)	Holds LOW for half period, then LOW→HIGH at mid-point
4	1	HIGH (end of period 3)	No — stays HIGH	1 (mid only)	Holds HIGH for half period, then HIGH→LOW at mid-point
5	0	LOW (end of period 4)	Yes — LOW→HIGH	2 (start + mid)	Jumps HIGH immediately, holds to mid-point, then HIGH→LOW
6	0	LOW (end of period 5)	Yes — LOW→HIGH	2 (start + mid)	Jumps HIGH immediately, holds to mid-point, then HIGH→LOW
7	1	LOW (end of period 6)	No — stays LOW	1 (mid only)	Holds LOW for half period, then LOW→HIGH at mid-point

Counting shortcut: Rather than tracking the absolute signal level, the receiver simply counts transitions per bit period. One transition → bit = 1. Two transitions → bit = 0. This rule holds regardless of the signal polarity or the starting state, confirming that no initialisation handshake is required between sender and receiver.

Reading the decode table directly: wherever you see "yes" for a start transition, that bit is a 0. Wherever you see "no", that bit is a 1. The mid-bit transitions (amber circles) are present in every single period — they provide the clock but carry no data.

A useful shortcut emerges: count transitions per period.

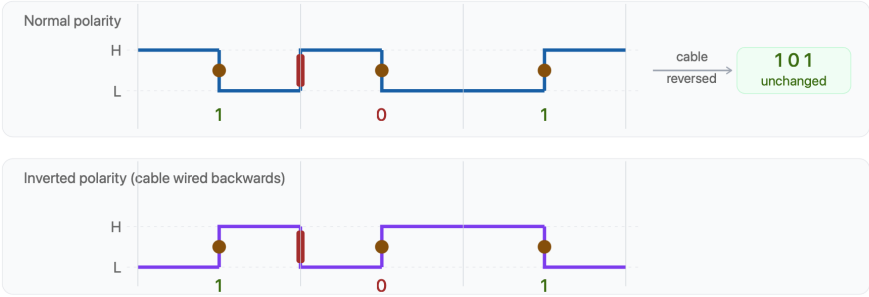
One transition = 1. Two transitions = 0.

Differential Manchester Encoding — Polarity Independence

Why reversing the cable polarity does not corrupt the decoded data

The differential property: In DME, bit values are encoded in the *presence or absence of a transition* at the start of each period — not in the absolute signal level. This means the receiver never needs to know which wire carries the logical HIGH. If the cable is wired backwards, the waveform is inverted but the transition pattern is identical, and every bit is decoded correctly without any re-configuration.

Polarity independence — the differential property



The transition pattern is identical in both waveforms — same positions, same count per period.
The receiver decodes 1 0 1 correctly regardless of polarity. No configuration needed.

What changes when polarity is inverted

The absolute signal levels are flipped — what was `HIGH` becomes `LOW` and vice versa. The waveform shape looks like a mirror image of the original. An oscilloscope on the inverted cable shows a different voltage pattern.

What does not change

The *position* of every transition stays exactly the same. The *count* of transitions per period stays exactly the same. The *presence or absence* of a start-of-period transition — the data encoding — stays exactly the same.

NRZ — polarity dependent

If you reverse an NRZ cable, every bit flips: all 1s become 0s and all 0s become 1s. The data is completely corrupted. The receiver has no way to detect or correct this without an out-of-band signal or a known preamble.

DME — polarity independent

Reversing a DME cable has no effect on the decoded data. The receiver decodes by counting transitions per period, which is unaffected by inversion. No preamble, no handshake, and no configuration change is needed.

Practical consequence: In systems using DME (token ring, ISDN, some industrial fieldbuses), the installer does not need to identify which terminal of a twisted pair is the positive and which is the negative conductor. Either orientation works. This simplifies cabling, eliminates a class of installation errors, and means a cable fault that reverses polarity does not silently corrupt data — the link continues operating correctly.

Interactive encoder — try your own bit sequence

Interactive Encoder

The four key properties — why DME is used

Property	How DME achieves it
Self-clocking	The mandatory mid-bit transition in every period gives the receiver a clock edge to lock onto, regardless of the data content
Polarity independence	Bit values are encoded in the *presence or absence* of a transition, not in signal level — inverting the signal does not change the decoded bits
DC balance	Every bit period contains at least one transition, ensuring equal time at HIGH and LOW over any sequence — no DC component accumulates
Bounded transitions	A 1 produces one transition per period; a 0 produces two. The maximum transition frequency (all-zeros sequence) is exactly twice the bit rate, which bounds the required signal bandwidth

DME is used in token ring networks (IEEE 802.4 and 802.5), ISDN, and various industrial fieldbuses.

In the context of the automotive Ethernet materials you have been studying, understanding DME also gives intuition for why 100BASE-T1 and 10BASE-T1S moved to more bandwidth-efficient PAM3 line codes — PAM3 encodes more bits per baud and eliminates the factor-of-two bandwidth overhead that differential encodings like DME carry.